THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL MCQ TEST SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. $0.\overline{18} \times 11$ equals: 1

(a) 2 **(b)** 3 **(c)** 1 **(d)** $\frac{2}{3}$

2. $8x^3 + 27$ factorises as: 1

(a) $(2x+3)(4x^2+6x+9)$ **(b)** $(2x+3)^3$ **(c)** $(2x+3)(4x^2-6x+9)$ **(d)** $(2x-3)(4x^2+6x+9)$

3. If two chords of a circle bisect each other, they are: 1

(a) both diameters **(b)** parallel

(c) unequal **(d)** perpendicular to each other

4. If the radius of a circle is doubled, its area becomes: 1

(a) three times **(b)** eight times **(c)** two times **(d)** four times

5. If $P(E) = 0.35$ and $P(F) = 0.45$ and E, F cannot occur together, then $P(E \text{ or } F) =$ 1

(a) 0.20 **(b)** 0.80 **(c)** 0.10 **(d)** 0.1575

6. The nth term of an AP is $t_n = 3n + 2$. The sum of its first 16 terms is: 1

(a) 456 **(b)** 440 **(c)** 437 **(d)** 443

7. The smallest natural number with exactly 6 factors (counting 1 and itself) is: 1

(a) 10 **(b)** 13 **(c)** 12 **(d)** 16

8. The average marks of 72 students is 90 and of another 144 students is 45. The average of all 216 students is: 1

(a) 59 **(b)** 62 **(c)** 61 **(d)** 60

9. By Euclid's algorithm, gcd$(2624, 1824)$ is: 1

(a) 33 **(b)** 32 **(c)** 34 **(d)** 31

10. The diagonals of a quadrilateral are 54 cm and 77 cm. The perimeter of the parallelogram formed by joining the midpoints of its sides is: 1

(a) 264 cm **(b)** 131 cm **(c)** 393 cm **(d)** 196 cm

11. The slope of the line through $(12, 22)$ and $(-24, 23)$ is: 1

(a) $\frac{35}{36}$ **(b)** $-\frac{37}{36}$ **(c)** $-\frac{1}{36}$ **(d)** $\frac{71}{36}$

12. Take $\pi = \frac{22}{7}$. The volume of a cylinder of radius 35 cm and height 38 cm is: 1

(a) 131669 cm^3 **(b)** 175562 cm^3 **(c)** 146300 cm^3 **(d)** 160931 cm^3

13. The area of the triangle with vertices $(0, 0)$, $(18, 0)$ and $(0, 8)$ is: 1

(a) 74 **(b)** 73 **(c)** 71 **(d)** 72

14. The area (in square units) of the triangle formed by the line $y = 4x + 12$ and the two axes is: 1

(a) 20 **(b)** 19 **(c)** 18 **(d)** 17

15. The value of $0.\overline{4} + 0.\overline{8}$ is: 1

(a) 1 **(b)** $\frac{5}{3}$ **(c)** $\frac{4}{3}$ **(d)** $\frac{4}{5}$

16. If $x + y = 7$ and $xy = 10$, then $x^3 + y^3 =$ 1

(a) 132 **(b)** 134 **(c)** 135 **(d)** 133

17. Parallel chords of lengths 30 cm and 40 cm lie on opposite sides of the centre of a circle of radius 25 cm. The distance between them (in cm) is: 1

(a) 34 **(b)** 36 **(c)** 35 **(d)** 37

18. The area of a cyclic quadrilateral with sides 5 cm, 8 cm, 19 cm and 20 cm is (cm^2): 1

(a) 127 **(b)** 125 **(c)** 128 **(d)** 126

19. Two dice are rolled together. The probability that the product of the numbers is 6 is: 1

(a) $\frac{2}{9}$ **(b)** $\frac{8}{9}$ **(c)** $\frac{1}{9}$ **(d)** $\frac{1}{10}$

20. The 5th term of an AP is 44 and the 12th term is 16. The 17th term is: 1

(a) -4 **(b)** -12 **(c)** -8 **(d)** 0

21. The number of perfect squares between 250 and 2500 (both excluded) is: 1

(a) 35 **(b)** 34 **(c)** 36 **(d)** 33

22. A group of 51 people has average weight 102 kg. Another group has average weight 98 kg. Together the average weight is 100 kg. The number of people in the second group is: 1

(a) 53 **(b)** 52 **(c)** 51 **(d)** 50

23. Start with $n = 23$. Repeat: if n is even replace it by $n/2$, otherwise by $3n + 1$. The number of steps needed to reach 1 is: 1

(a) 17 **(b)** 14 **(c)** 16 **(d)** 15

24. The four angles of a quadrilateral are in AP with common difference 26°. The smallest angle is: 1

(a) 61° **(b)** 77° **(c)** 51° **(d)** 25°

25. The line $24x + 11y = 264$ cuts the axes at two points. The area of the triangle it forms with the axes (square units) is: 1

(a) 133 **(b)** 132 **(c)** 134 **(d)** 131

26. The diagonal of a cuboid measuring 4 cm $\times$ 8 cm $\times$ 8 cm is: 1

(a) 11 cm **(b)** 12 cm **(c)** 14 cm **(d)** 13 cm

27. The distance of the circumcentre of the right triangle with vertices $(0, 0)$, $(10, 0)$ and $(0, 24)$ from the origin is: 1

(a) 15 **(b)** 13 **(c)** 14 **(d)** 12

28. The x-coordinate of the point where $y = 7x - 2$ and $y = 3x + 18$ meet is: 1

(a) 4 **(b)** 7 **(c)** 6 **(d)** 5

29. $0.\overline{26} + 0.\overline{62}$ equals: 1

(a) $\frac{4}{5}$ **(b)** $\frac{8}{9}$ **(c)** 1 **(d)** $\frac{7}{9}$

30. If $x - y = 3$ and $xy = 10$, then $x^3 - y^3 =$ 1

(a) 118 **(b)** 117 **(c)** 116 **(d)** 119

31. Parallel chords of lengths 30 cm and 40 cm lie on the same side of the centre of a circle of radius 25 cm. The distance between them (in cm) is: 1

(a) 6 **(b)** 5 **(c)** 7 **(d)** 4

32. A circle of radius 14 cm (take $\pi = \frac{22}{7}$) is inscribed in a square. The area of the square not covered by the circle is (cm^2): 1

(a) 167 **(b)** 169 **(c)** 170 **(d)** 168

33. Two dice are rolled together. The probability that the sum is a prime number is: 1

(a) $\frac{5}{12}$ **(b)** $\frac{5}{13}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{7}{12}$

34. The sum of the first n terms of a sequence is $S_n = 3n^2 + 1n$. Its 12th term is: 1

(a) 73 **(b)** 67 **(c)** 70 **(d)** 76

35. For the expression $n^2 - n + 17$, the smallest natural number n for which the value is NOT prime is: 1

(a) 16 **(b)** 19 **(c)** 18 **(d)** 17

36. In what ratio must a 34% solution and a 72% solution be mixed to obtain a 60% solution? 1

(a) $6 : 14$ **(b)** $7 : 13$ **(c)** $13 : 6$ **(d)** $6 : 13$

37. Using Euclid's subtraction algorithm for gcd$(1971, 1251)$, the number of subtraction steps needed is: 1

(a) 15 **(b)** 16 **(c)** 13 **(d)** 14

38. The sides of a rectangle are in the ratio $3 : 4$ and its diagonal is 10 cm. Its perimeter is: 1

(a) 42 cm **(b)** 28 cm **(c)** 30 cm **(d)** 38 cm

39. A taxi fare is ₹300 for 4 km and ₹600 for 10 km (fare is a linear function of distance). The fare for 29 km (in ₹) is: 1

(a) 1500 **(b)** 1600 **(c)** 1550 **(d)** 1650

40. Take $\pi = \frac{22}{7}$. A toy is a cone of height 12 cm placed on a hemisphere, both of radius 21 cm. Its total volume is: 1

(a) 29939 cm^3 **(b)** 22453 cm^3 **(c)** 24948 cm^3 **(d)** 27443 cm^3

41. **Assertion (A):** The probability of an impossible event is 0. 1

Reason (R): The probability of any event lies between 0 and 1, both included.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

42. **Assertion (A):** The sequence $2, 4, 8, 16, \ldots$ is a geometric progression. 1

Reason (R): The sequence $1, 3, 5, 7, \ldots$ is an arithmetic progression.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

43. **Assertion (A):** The statement 'Every prime number is odd' is false. 1

Reason (R): The prime number 2 is not odd.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

44. **Assertion (A):** The combined average of two groups is always the simple mean of the two group averages. 1

Reason (R): The combined average is the total of all the values divided by the total number of values.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

45. **Assertion (A):** $29 \bmod 7 = 1$. 1
Reason (R): $7 \bmod 29 = 7$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** The diagonals of a parallelogram bisect each other. 1
Reason (R): The diagonals of a parallelogram are always equal.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** A linear equation in two variables has infinitely many solutions. 1
Reason (R): The graph of a linear equation in two variables is a straight line.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** The total surface area of a cube of side 4 cm is 96 cm^2. 1
Reason (R): The lateral surface area of a cube of side a is $4a^2$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** The point $(4, -2)$ lies in Quadrant IV. 1
Reason (R): In Quadrant IV both the abscissa and the ordinate are negative.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** The slope of the line $y = -4x + 7$ is -4. 1
Reason (R): For the line $y = ax + b$ the slope is b.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**